

Workshop on Understanding and Mitigating Cognitive Biases in Human-AI Collaboration

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ABSTRACT

AI systems are increasingly incorporated into human decision-making. Yet, human decision-makers are often affected by their cognitive biases. In critical settings, such as medical diagnosis, criminal judgment, or information consumption, these cognitive biases hinder optimal decision outcomes, thereby resulting in dangerous decisions and negative societal impact. The use of AI systems can amplify and exacerbate cognitive biases in their users. In this workshop, we seek to foster discussions on ongoing research around cognitive biases in human-AI collaboration and identify future research directions to understand, quantify, and mitigate the effects of cognitive biases. We will explore cognitive biases appearing in various contexts of human-AI collaboration: what can cause them?; how can we measure, model, mitigate, and manage cognitive biases?; and how can we utilise cognitive biases for the greater good? We will reflect on workshop discussions to form a research community around cognitive biases and bias-aware systems.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Collaborative and social computing.**

KEYWORDS

Cognitive Bias, Human-AI Collaboration, Debiasing

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1 INTRODUCTION

In recent years, Artificial Intelligence (AI) has been increasingly outperforming humans in many tasks, such as classification and forecasting [18, 38]. We have seen a rapid uptake in the deployment of AI systems to complement and support human decision-makers in critical domains: judges use algorithmic risk assessment to determine criminal sentences, doctors rely on machine learning models to diagnose patients, and online media platforms adopt recommendation systems to present users with relevant content items. However, the literature indicates that human decision-makers are not always rational [45]; their decisions are often affected by *cognitive biases* – defined by Tversky and Kahneman [50] as mental shortcuts or heuristics to make faster but less deliberate decisions. Cognitive biases distort our thinking in a way we are often unaware of and can negatively influence decision outcomes. For example, confirmation bias can affect how users interpret and seek information online [1], anchoring bias can induce unfair juridical decisions when presented with multiple pieces of evidence [20], and the Dunning-Kruger effect can hinder appropriate reliance on AI systems [21].

Research has suggested that AI systems can trigger and even amplify cognitive biases in their users [1, 3, 7, 33, 35]. Personalised recommendation systems, for example, optimise content recommendations around the users' preferences and cater predominantly to what users prefer. As a result, such systems risk reinforcing confirmation bias and the echo chamber effect [1, 6, 26]. Moreover, studies have shown that AI explanations can exacerbate our cognitive biases and compromise AI-assisted decision-making, such as trust in AI [33], reliance on AI [5, 10, 21, 39], and interpretation of AI results [27, 46]. Meanwhile, cognitive biases can shape the quality of

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ground-truth data and thereby influence downstream applications and human evaluations of systems [16, 24], and also influence the outcomes of AI systems [1, 4]. Recommendation systems pick up not only user preferences but also their confirmation bias through their selective information consumption behaviour. As a result, these systems deliver content that, in turn, amplifies users' cognitive biases [1]. A recent example of ChatGPT also demonstrates that it exhibits many biases humans possess, for instance, framing bias and overconfidence bias [12]. With AI systems and cognitive biases forming an interplay that influences human decision-making, it is, therefore, crucial to understand how cognitive biases manifest themselves and how their effects can be mitigated.

A growing body of work has explored how to mitigate cognitive biases in human-AI collaboration. By leveraging AI systems and carefully designing them as a decision aid, Kliegr et al. [27] reviewed twenty cognitive biases in interpreting rule-based machine learning models and associated debiasing techniques. Wang et al. [51] proposed a framework for building explainable AI systems that reduce common cognitive biases, e.g., availability bias and confirmation bias. In addition, some research has proposed technological interventions to counter cognitive biases in human-AI collaboration. Bućinca et al. [10] employed cognitive forcing tools to reduce overreliance on AI. Rastogi et al. [43] introduced a time allocation strategy to mitigate anchoring bias. Bach et al. [3] further evaluated bias mitigation techniques as integrated into the user interface of a clinical decision support tool and identified concerns around the impact on efficiency.

Spanning from literature in behavioural economics, researchers and practitioners have adopted *nudges* as an intervention to counter the undesired effects of cognitive biases [9, 11, 31, 36, 37]. Nudges alter the environment, i.e., the user interface, and subsequently trigger cognitive biases that shift users towards a particular decision or behaviour [48]. Therefore, such methods not only leverage cognitive biases but can also be used to combat their negative effects. For instance, Rieger et al. [44] employed targeted obfuscation on search results to nudge people towards decreasing interaction with attitude-confirming information. By obfuscating content items that may confirm one's beliefs, this nudge taps into the status-quo bias – a tendency to go along with the path of least resistance – and, in turn, helps reduce confirmation bias as users avoid interacting with the obfuscated items.

Nonetheless, effectively mitigating cognitive biases is a challenging task, particularly because of the inherent property of cognitive biases that some individuals are less or more susceptible to biased judgments due to interaction contexts (e.g., domain knowledge, cognitive load, or topic involvement) [8, 30, 35] and individual characteristics (e.g., short-term memory span) [35, 42]. Moreover, different individuals, such as those with varying levels of expertise, possess different mental models of interacting with and understanding AI systems [25, 53]. As a result, not every debiasing intervention would always be effective for every user [21, 35, 44]. Additionally, multiple cognitive biases can manifest at the same time, producing mixed effects that can be difficult to observe and mitigate [2].

The role of cognitive biases in human-AI collaboration has become a growing discourse in the CSCW community. Through different domains, such as HCI, CSCW, Information Retrieval, and

Behavioural Economics, diverse forms of studies address the question of how cognitive biases manifest themselves and how they could be effectively mitigated. Therefore, it is important to bridge together insights from different disciplines and create a common ground for future cognitive bias research.

In this workshop, we aim to bring together researchers, practitioners, and designers to jointly seek a better understanding of cognitive biases and solutions to mitigate problems arising from biases. Recent workshops such as *Workshop on Detection and Design for Cognitive Biases in People and Computing Systems* [13] at CHI 2020, *Workshop on Technologies to Support Critical Thinking in an Age of Misinformation* [14] at CHI 2021, and *the Dagstuhl Seminar on Technologies to Support Critical Thinking in an Age of Misinformation* [15] have explored related topics with particular focus on online information consumption and misinformation. This proposed workshop at CSCW 2023 will focus on cognitive biases in the context of human-AI collaboration, where AI systems act as supporting tools for human decision-makers.

2 WORKSHOP GOALS AND THEMES

We aim to foster a discussion about ongoing work on cognitive biases in HCI, provide a common platform to revisit the current research, and establish a research agenda for understanding, quantifying, mitigating, and utilising cognitive biases. Ultimately, we seek to form a research community that works towards the design of *Bias-Aware Systems* [8, 34], defined as computing systems that take into account the cognitive biases of their users. Through creating this community, we aim to build a shared understanding of cognitive biases and methods to measure, utilise, and mitigate their effects. We hope that discussions in this workshop lead to fruitful collaborations that leverage our understanding of cognitive biases in users.

In this workshop, we call for participants to share their research ideas, questions, and opinions with regard to the following themes:

- **Discovering and Identifying Cognitive Biases** We would like to explore mechanisms and components of AI systems that amplify or trigger cognitive biases in their users. In what human-AI collaboration scenarios are cognitive biases involved? Recent research has explored a diverse set of cognitive biases people follow when interacting with explainable AI systems [7, 17, 27].
- **Modelling and Quantifying Cognitive Biases:** An important step towards bias mitigation is to model cognitive biases and measure their extent [4, 40]. However, since users are often unaware of their cognitive biases, it is challenging to know whether cognitive biases are manifesting themselves. Recent research has proposed mathematical frameworks to model cognitive biases [23, 35, 43]. Moreover, some works have explored methods to reliably quantify cognitive biases in-situ using a variety of physiological sensors [8, 19].
- **Novel Approaches to Mitigate Cognitive Biases:** We would like to explore novel methods to mitigate the negative effects of cognitive biases in human-AI collaboration. Existing approaches include *nudging*, i.e., changing the choice environment [11], *boosting*, i.e., fostering metacognitive skills in people [28], and designing *decision support systems* that help users make effective and accurate decisions [51]. We seek to discuss the shortcomings and

limitations of existing debiasing approaches and develop future directions.

- **Application Scenarios of Cognitive Biases:** While it is known that cognitive biases negatively affect human decision-making, we would like to explore the use of cognitive biases for the greater good. Can we imagine scenarios in which cognitive biases actually benefit human-AI collaboration? [29, 32]
- **Impact of the Bias Mitigation:** We seek to explore how bias identification and mitigation strategies can positively and negatively impact AI systems and their users. What benefits do people get if their biases are mitigated? Do we really need to eliminate biases? Is there an alternative way to support human decision-making? Recent research has shown that some debiasing interventions like nudges can harm user autonomy [3, 36] or slow down the interaction [41].
- **Case Studies of Cognitive Biases in Human-AI Collaboration:** Presentation of concrete cases where the prevalence, mitigation, and utilisation of cognitive biases in human-AI collaboration have been investigated.

3 CALL FOR PARTICIPATION

We would like to welcome 20-30 participants for this workshop (excluding the organisers). Participants will be required to contribute a brief statement of interest to the workshop. We accept several forms of submission, including (1) a short research summary or position paper (2-4 pages excluding references) discussing one or more workshop themes or (2) a one-page essay stating motivations for attending this workshop with a short bio. Each submission will be reviewed by the workshop organisers and accepted based on the quality of the submission and the diversity of perspectives to allow fruitful discussions between researchers from different domains, including but not limited to HCI, CSCW, AI, and cognitive psychology. Upon acceptance, we will encourage participants to record a short 3-5 minute video presenting the content of their submission, which will be available to watch before the workshop. We will advertise our workshop and the call for papers through mailing lists, social media, and forums.

4 WORKSHOP SCHEDULE

We propose a one-day workshop with hybrid participation: there will be an option to participate physically at CSCW 2023 and virtually to ensure maximum inclusion. We plan to organise the workshop with the following activities:

- **Introduction** (1 hour): We will welcome participants to this workshop and provide an outline of planned activities, goals, and themes. We will also include a quick ice-breaking activity for participants to get to know each other.
- **Lighting Talks** (1.5 hours): Participants will share their paper submissions. We plan on allocating time for selected papers, and the presentations will be organised under workshop themes. Each presenter will have three minutes to talk about their work and two minutes for Q&A. We aim for this session to be an opportunity for authors to introduce their research and gain feedback from the audience.
- **Two-round Action Group Activities** (2 hours): We will divide participants into *action groups* where each group's theme will

associate with concrete scenarios from the submitted position papers and existing pre-workshop discussions, for example, recommender systems, explainable AI, or generative AI. Participants can join the group with the theme they are most interested in. The number of action groups will be determined prior to the workshop. In two rounds, participants will engage in the following activities:

- **Brainstorming** (40 minutes): Participants will be assigned a brainstorming task and discuss solutions in their action group. Each brainstorming task will be associated with one of the abovementioned themes. In both rounds, there will be at least one organiser facilitating discussion in each group. We plan to source brainstorming tasks from workshop themes: *what are cognitive biases and their causes-triggers-effects* (discovering bias); *what are the measures of cognitive biases* (quantifying bias); *develop interventions to debias* (mitigating bias); *identify application scenarios where cognitive biases are exploited* (utilising bias); and *identify positive and negative impacts from bias mitigation* (impact of bias mitigation).
- **Knowledge Synthesis** (20 minutes): All participants will reconvene to share what they discussed in their action groups, including key ideas, challenges, and opportunities.
- **Closing Remarks** (30 minutes): We will synthesise key takeaways from discussions and identify the next steps for building a research community on cognitive bias. We will also facilitate follow-up conversations after formally concluding the workshop.

We will incorporate breaks between sessions and social activities into the final schedule.

5 HYBRID SETUP

We plan to utilise the following tools to support and accommodate our hybrid setup:

- **Workshop Website.** We will make the workshop information publicly available on the workshop website¹, including the workshop proposal, call for participation, accepted submissions, workshop program, participant information, and other relevant material.
- **Slack Workspace.** We will set up a dedicated Slack workspace to enable asynchronous communication among workshop participants before, during, and after the workshop. Prior to the workshop, we will also share accepted paper submissions and (if applicable) short videos on the Slack workspace. The organisers will actively monitor discussions on the channels to keep our participants engaged.
- **Zoom Video Conferencing.** We will broadcast in-person workshop presentations, activities, and discussions on Zoom to allow virtual participants to take part in our workshop. The same tool will also allow live closed-captioning to support accessibility.
- **Padlet and Miro Boards.** We will use Padlet and Miro boards to allow participants, both in-person and virtual, to share and note down ideas throughout the workshop. We will encourage workshop participants to take notes on these online sharing tools as they will be accessible for remote participants and future references.

¹<http://www.critical-media.org/cscw23>

6 WORKSHOP OUTCOMES

The following are the expected outcomes of this workshop.

- **Forming a research community.** By bringing together researchers and practitioners from different disciplines, we expect to see an exchange of knowledge and future collaborations on research around cognitive biases and bias-aware systems. We will also keep the slack workspace open to workshop participants to continue to develop a community of cognitive bias researchers.
- **Compilation of Cognitive Biases in HCI.** Based on workshop discussions, we intend to document a list of cognitive biases in various HCI contexts and the associated quantification, utilisation, and mitigation strategies, accompanied by empirical studies that explored such biases.
- **Sharing Insights.** We will document the results of the action groups and discussions and make the collected information available to workshop participants and the broader community through an online repository and a public website.

7 ORGANISERS

The organising team of this workshop consists of researchers and experts working in and across CSCW, HCI, AI, and Information Retrieval.

- **Nattapat Boonprakong** is a PhD candidate at the School of Computing and Information Systems, the University of Melbourne. He is interested in Cognition-aware Systems and Empathic Computing. His PhD research is particularly focused on the detection and mitigation of cognitive biases in online information consumption where people often face different opinions.
- **Gaole He** is a PhD candidate at Web Information Systems group of the Faculty of Electrical Engineering, Mathematics and Computer Science (EEMCS/EWI), Delft University of Technology. His research focuses on human-centered explainable AI, human-AI decision-making, and knowledge graph reasoning.
- **Ujwal Gadiraju** is an Assistant Professor at the Software Technology Department of the EEMCS faculty at Delft University of Technology. He is a Co-director of the TU Delft Design@Scale AI Lab. His research focuses on *Human-Centered AI and Crowd Computing* to create novel methods, interfaces, systems, and tools capable of overcoming existing challenges at the intersection of HCI and AI in our pursuit to build better AI systems and facilitate better reliance of humans on AI systems. He has (co)-organized workshops and symposiums focused on biases in human computation, crowdsourcing, and human-AI interactions.
- **Niels van Berkel** is an Associate Professor at the Department of Computer Science at Aalborg University. His research focuses on collaboration with real-world intelligent systems, with a focus on overcoming challenges in decision-making. He has (co)-organised various workshops, including on the topic of Human-AI interaction, explainability, and fairness [52].
- **Danding Wang** is an Assistant Researcher at the Institute of Computing Technology, Chinese Academy of Sciences. Her research focuses on human-centered explainable AI, AIGC detection, media forensics, and misinformation detection. She proposed an explainable AI framework that facilitates human reasoning and mitigates cognitive biases [51].

- **Si Chen** is a PhD candidate at the School of Information Sciences at University of Illinois at Urbana-Champaign. She conducts research on human consciousness and metacognition while using AI-driven intelligent learning systems. Additionally, she explores how to ensure inclusivity in such systems for learners with diverse abilities, such as those who are blind and visually impaired.
- **Jiqun Liu** is an Assistant Professor of Data Science and Affiliated Assistant Professor of Psychology at the University of Oklahoma (OU). His research focuses on the intersection of human-centered data science, interactive information seeking/retrieval, and cognitive psychology, and seeks to apply the knowledge learned about people interacting with information in adaptive recommendation, user education and intelligent nudging.
- **Benjamin Tag** is a Lecturer in the Human-Centred Computing Group at Monash University. He researches Human-AI Interaction, Digital Emotion Regulation, and Immersive Analytics with a special focus on inferring mental state changes from data collected in the wild. Benjamin co-organized a series of workshops on cognitive biases [13–15].
- **Jorge Goncalves** is an Associate Professor in the School of Computing and Information Systems at the University of Melbourne. He has conducted extensive research on Human Computation and facilitating Human-AI Interaction. He has also served as Workshops Co-Chair for CHI'19 and CHI'20, and co-organised many successful workshops at leading HCI venues such as CHI, CSCW and Ubicomp [22, 47, 49].
- **Tilman Dingler** is a Senior Lecturer in the School of Computing and Information Systems at the University of Melbourne. He investigates the notion of cognition-aware systems and builds technologies that support users' information-processing capabilities. Tilman instigated a series of recent workshops related to critical thinking and the role of cognitive biases at prime venues, such as CHI and in the prestigious Dagstuhl seminar series [13–15].

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